

1.5 Elementary Matrices and a Method for Finding A^{-1}

This section develops an algorithm for finding the inverse of a matrix, and along the way establishes further properties of invertible matrices.

Elementary Matrices

Recall the three elementary row operations on a matrix A (Section 1.1): multiply a row by a nonzero constant c ; interchange two rows; add c times one row to another. Each has an **inverse operation** that undoes it: multiply the same row by $1/c$; interchange the same two rows back; if c times row r_i was added to row r_j , add $-c$ times r_i to r_j . Hence if B results from A by a sequence of row operations, some second sequence restores A from B .

Definition 1

Matrices A, B are **row equivalent** if either (hence each) can be obtained from the other by a sequence of elementary row operations.

Next we show how matrix multiplication can perform a row operation.

Definition 2

A matrix E is an **elementary matrix** if it results from I by a *single* elementary row operation.

EXAMPLE 1 | Elementary Matrices and Row Operations

$$\begin{bmatrix} 1 & 0 \\ 0 & -3 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

result from, respectively: multiplying row 2 of I_2 by -3 ; interchanging rows 2, 4 of I_4 ; adding 3 times row 3 of I_3 to row 1; multiplying row 1 of I_3 by 1.

Theorem 1.5.1 – Row Operations by Matrix Multiplication

If elementary matrix E results from performing a row operation on I_m , and A is $m \times n$, then EA is the matrix that results from performing that same row operation on A .

EXAMPLE 2 | Using Elementary Matrices

For $A = \begin{bmatrix} 1 & 0 & 2 & 3 \\ 2 & -1 & 3 & 6 \\ 1 & 4 & 4 & 0 \end{bmatrix}$ and $E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix}$ (which adds 3 times row 1 of I_3 to row 3):

$$EA = \begin{bmatrix} 1 & 0 & 2 & 3 \\ 2 & -1 & 3 & 6 \\ 4 & 4 & 10 & 9 \end{bmatrix}$$

exactly the result of adding 3 times row 1 of A to row 3.

Table 1 lists each row operation that produces E from I , alongside its **inverse operation**, which restores I from E .

Row Operation on I That Produces E	Row Operation on E That Reproduces I
Multiply row i by $c \neq 0$	Multiply row i by $1/c$
Interchange rows i and j	Interchange rows i and j
Add c times row i to row j	Add $-c$ times row i to row j

EXAMPLE 3 | Row Operations and Inverse Row Operations

$$\begin{aligned} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} &\xrightarrow{\text{multiply row 2 by 7}} \begin{bmatrix} 1 & 0 \\ 0 & 7 \end{bmatrix} \xrightarrow{\text{multiply row 2 by } \frac{1}{7}} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} &\xrightarrow{\text{interchange rows 1, 2}} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \xrightarrow{\text{interchange rows 1, 2}} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} &\xrightarrow{\text{add } 5 \times \text{row 2 to row 1}} \begin{bmatrix} 1 & 5 \\ 0 & 1 \end{bmatrix} \xrightarrow{\text{add } -5 \times \text{row 2 to row 1}} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \end{aligned}$$

Theorem 1.5.2

Every elementary matrix is invertible, and its inverse is also an elementary matrix.

Proof Let E result from a row operation on I , and let E_0 result from applying the *inverse* operation to I . By Theorem 1.5.1 and the fact that inverse row operations cancel, $E_0E = I$ and $EE_0 = I$; so $E_0 = E^{-1}$. ■

Equivalence Theorem

Linear algebra's diverse ideas are deeply connected. The following theorem links invertibility, homogeneous systems, reduced row echelon form, and elementary matrices – a first step in that direction (more will be added later).

Theorem 1.5.3 – Equivalent Statements

For an $n \times n$ matrix A , the following are equivalent (all true or all false):

- (a) A is invertible.
- (b) $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.
- (c) The reduced row echelon form of A is I_n .
- (d) A is a product of elementary matrices.

Proof We show $(a) \Rightarrow (b) \Rightarrow (c) \Rightarrow (d) \Rightarrow (a)$, which establishes that all four are equivalent.

$(a) \Rightarrow (b)$: If A is invertible and $A\mathbf{x}_0 = \mathbf{0}$, multiplying by A^{-1} gives $\mathbf{x}_0 = A^{-1}\mathbf{0} = \mathbf{0}$ – only the trivial solution.

$(b) \Rightarrow (c)$: If $A\mathbf{x} = \mathbf{0}$ has only the trivial solution, Gauss–Jordan elimination reduces the augmented matrix $[A \mid \mathbf{0}]$ to $[I_n \mid \mathbf{0}]$ (each $x_i = 0$); disregarding the zero last column, the reduced row echelon form of A is I_n .

$(c) \Rightarrow (d)$: If A reduces to I_n by elementary row operations, then by Theorem 1.5.1 there are elementary matrices $E_1, \dots, E_k$ with $E_k \cdots E_1 A = I_n$. Each E_i is invertible (Theorem 1.5.2), so $A = E_1^{-1} \cdots E_k^{-1}$ – a product of elementary matrices.

$(d) \Rightarrow (a)$: A product of elementary matrices is a product of invertible matrices (Theorem 1.5.2), hence invertible by Theorem 1.4.6. ■

A Method for Inverting Matrices

If A is invertible, the elementary matrices reducing A to I_n satisfy $E_k \cdots E_1 A = I_n$; multiplying on the right by A^{-1} gives $A^{-1} = E_k \cdots E_1 I_n$. That is, *the same row operations that reduce A to I_n transform I_n into A^{-1} .*

Inversion Algorithm

To find A^{-1} : find a sequence of elementary row operations reducing A to I , and apply that same sequence to I to obtain A^{-1} .

In practice, adjoin I to the right of A to form $[A \mid I]$, then row-reduce until the left block is I ; the right block is then A^{-1} , giving $[I \mid A^{-1}]$.

EXAMPLE 4 | Using Row Operations to Find A^{-1}

Find the inverse of $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$.

SOLUTION:

$$\begin{aligned} & \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 5 & 3 & 0 & 1 & 0 \\ 1 & 0 & 8 & 0 & 0 & 1 \end{array} \right] \xrightarrow{\substack{-2R_1+R_2 \\ -1R_1+R_3}} \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & -2 & 5 & -1 & 0 & 1 \end{array} \right] \\ & \xrightarrow{2R_2+R_3} \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & -1 & -5 & 2 & 1 \end{array} \right] \xrightarrow{-1 \times R_3} \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array} \right] \end{aligned}$$

$$\xrightarrow[-3R_3+R_1]{3R_3+R_2} \left[\begin{array}{ccc|ccc} 1 & 2 & 0 & -14 & 6 & 3 \\ 0 & 1 & 0 & 13 & -5 & -3 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array} \right] \xrightarrow{-2R_2+R_1} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -40 & 16 & 9 \\ 0 & 1 & 0 & 13 & -5 & -3 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array} \right]$$

Thus, $A^{-1} = \begin{bmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix}$.

If A is not invertible, this will show up as a row of zeros appearing on the *left* block at some stage – by Theorem 1.5.3(a),(c), A cannot then be invertible, and the computation may stop.

EXAMPLE 5 | Showing That a Matrix Is Not Invertible

For $A = \begin{bmatrix} 1 & 6 & 4 \\ 2 & 4 & -1 \\ -1 & 2 & 5 \end{bmatrix}$:

$$\left[\begin{array}{ccc|ccc} 1 & 6 & 4 & 1 & 0 & 0 \\ 2 & 4 & -1 & 0 & 1 & 0 \\ -1 & 2 & 5 & 0 & 0 & 1 \end{array} \right] \xrightarrow[R_1+R_3]{-2R_1+R_2} \left[\begin{array}{ccc|ccc} 1 & 6 & 4 & 1 & 0 & 0 \\ 0 & -8 & -9 & -2 & 1 & 0 \\ 0 & 8 & 9 & 1 & 0 & 1 \end{array} \right] \xrightarrow{R_2+R_3} \left[\begin{array}{ccc|ccc} 1 & 6 & 4 & 1 & 0 & 0 \\ 0 & -8 & -9 & -2 & 1 & 0 \\ 0 & 0 & 0 & -1 & 1 & 1 \end{array} \right]$$

A zero row appears on the left, so A is not invertible.

EXAMPLE 6 | Analyzing Homogeneous Systems

Determine whether each has nontrivial solutions: (a) $x_1 + 2x_2 + 3x_3 = 0$, $2x_1 + 5x_2 + 3x_3 = 0$, $x_1 + 8x_3 = 0$
 (b) $x_1 + 6x_2 + 4x_3 = 0$, $2x_1 + 4x_2 - x_3 = 0$, $-x_1 + 2x_2 + 5x_3 = 0$

SOLUTION: By Theorem 1.5.3(a),(b), a homogeneous system has only the trivial solution iff its coefficient matrix is invertible. The coefficient matrix of (a) is the A of Example 4, which is invertible, so (a) has only the trivial solution; the coefficient matrix of (b) is the A of Example 5, which is not invertible, so (b) has nontrivial solutions.

Exercise Set 1.5

1. Determine whether the matrix is elementary.

(a) $\begin{bmatrix} 1 & 0 \\ -5 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} -5 & 1 \\ 1 & 0 \end{bmatrix}$ (c) $\begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ (d) $\begin{bmatrix} 2 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

3. Find a row operation and its elementary matrix that restores the given elementary matrix to I .

(a) $\begin{bmatrix} 1 & -3 \\ 0 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} -7 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ (c) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -5 & 0 & 1 \end{bmatrix}$ (d) $\begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

5. Identify the row operation corresponding to E and verify that EA results from applying it to A .

(a) $E = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $A = \begin{bmatrix} -1 & -2 & 5 & -1 \\ 3 & -6 & -6 & -6 \end{bmatrix}$

$$(b) E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -3 & 1 \end{bmatrix}, A = \begin{bmatrix} 2 & -1 & 0 & -4 & -4 \\ 1 & -3 & -1 & 5 & 3 \\ 2 & 0 & 1 & 3 & -1 \end{bmatrix}$$

$$(c) E = \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, A = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$$

$$7. \text{ Using } A = \begin{bmatrix} 3 & 4 & 1 \\ 2 & -7 & -1 \\ 8 & 1 & 5 \end{bmatrix}, B = \begin{bmatrix} 8 & 1 & 5 \\ 2 & -7 & -1 \\ 3 & 4 & 1 \end{bmatrix}, C = \begin{bmatrix} 3 & 4 & 1 \\ 2 & -7 & -1 \\ 2 & -7 & 3 \end{bmatrix}, \text{ find an elementary matrix } E$$

satisfying:

$$(a) EA = B \quad (b) EB = A \quad (c) EA = C \quad (d) EC = A$$

9. First use Theorem 1.4.5, then the inversion algorithm, to find A^{-1} if it exists.

$$(a) A = \begin{bmatrix} 1 & 4 \\ 2 & 7 \end{bmatrix} \quad (b) A = \begin{bmatrix} 2 & -4 \\ -4 & 8 \end{bmatrix}$$

11. Use the inversion algorithm to find the inverse, if it exists.

$$(a) \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix} \quad (b) \begin{bmatrix} -1 & 3 & -4 \\ 2 & 4 & 1 \\ -4 & 2 & -9 \end{bmatrix}$$

$$13. \text{ Use the inversion algorithm to find the inverse, if it exists, of } \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}.$$

19. Find the inverse of each 4×4 matrix, where k_1, k_2, k_3, k_4, k are all nonzero.

$$(a) \begin{bmatrix} k_1 & 0 & 0 & 0 \\ 0 & k_2 & 0 & 0 \\ 0 & 0 & k_3 & 0 \\ 0 & 0 & 0 & k_4 \end{bmatrix} \quad (b) \begin{bmatrix} k & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & k & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

21. Find all values of c , if any, for which the matrix is invertible.

$$\begin{bmatrix} c & c & c \\ 1 & c & c \\ 1 & 1 & c \end{bmatrix}$$

23. Express the matrix and its inverse as products of elementary matrices.

$$\begin{bmatrix} -3 & 1 \\ 2 & 2 \end{bmatrix}$$

$$27. \text{ Show that } A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 4 & 1 \\ 2 & 1 & 9 \end{bmatrix} \text{ and } B = \begin{bmatrix} 1 & 0 & 5 \\ 0 & 2 & -2 \\ 1 & 1 & 4 \end{bmatrix} \text{ are row equivalent by finding a sequence of elementary row operations that produces } B \text{ from } A, \text{ and use it to find a matrix } C \text{ with } CA = B.$$

29. Show that if $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ a & b & c \end{bmatrix}$ is an elementary matrix, then at least one entry in the third row must be zero.

30. Show that

$$A = \begin{bmatrix} 0 & a & 0 & 0 & 0 \\ b & 0 & c & 0 & 0 \\ 0 & d & 0 & e & 0 \\ 0 & 0 & f & 0 & g \\ 0 & 0 & 0 & h & 0 \end{bmatrix}$$

is not invertible for any values of the entries.

31. **(Working with Proofs)** Prove that if A, B are $m \times n$ matrices, then A, B are row equivalent if and only if they have the same reduced row echelon form.
32. **(Working with Proofs)** Prove that if A is invertible and B is row equivalent to A , then B is also invertible.
33. **(Working with Proofs)** Prove that if B is obtained from A by a sequence of elementary row operations, then a second sequence of elementary row operations, applied to B , recovers A .